Hay Fever

Allergic rhinitis is commonly known as hay fever. Allergic rhinitis develops when the body's immune system becomes sensitized and overreacts to something in the environment like pollen grains, strong odour of perfumes, dust etc. that typically causes no problem in most people. When a sensitive person inhales an allergen the body's immune system may react with the symptoms such as sneezing, cough and puffy swollen eyelids.

Type II Hypersensitivity: Antibody dependent hypersensitivity

In this type of hypersensitivity reactions the antibodies produced by the immune response binds to antigens on the patient's own cell surfaces. It is also known as cytotoxic hypersensitivity and may affect variety of organs or tissues. Ig G and Ig M antibodies bind to these antigens and form complexes. This inturn activates the classical complement pathway and eliminates the cells presenting the foreign antigen. The reaction takes hours to day (Figure 11.8).

Drug induced haemolytic anaemia

Certain drugs such as penicillin, cephalosporin and streptomycin can absorb non-specifically to protein on surface of RBC forming complex similar to hapten-carrier complex. In some patients these complex induce formation of antibodies, which binds to drugs on RBC and induce complement mediated lysis of RBC and thus produce progressive anaemia. This drug induced haemolytic anaemia is an example of Type II hypersensitivity reaction.

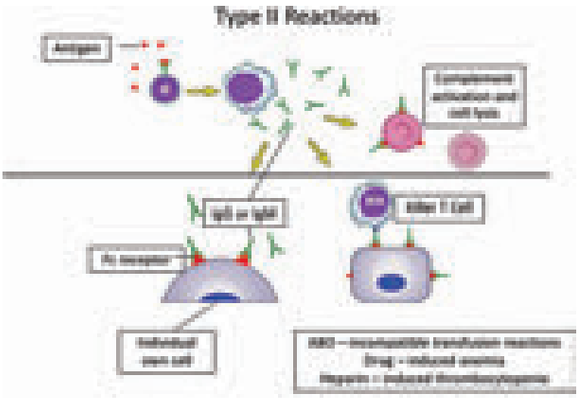


Figure 11.8: Type II hypersensitivity

Type III Hypersensitivity: Immune complex mediated hypersensitivity

When a huge amount of antigen enters into the body, the body produces higher concentrations of antibodies. These antigens and antibodies combine together to form insoluble complex called immune complex. These complexes are not completely removed by macrophages. These get attached to minute capillaries of tissues and organs such as kidneys, lung and skin (Figure 11.9). These antigen-antibody complexes activate the classical complement pathway leading to vasodilation. The complement proteins and antigen-antibody complexes attract leucocytes to the area. The leukocytes discharge their killing agents and promote massive inflammation. This can lead to tissue death and haemorrhage.

Arthus reaction

It was first observed by Arthus. It is a local immune complex reaction occurring in the skin. Horse serum and egg albumin are the antigens that induce the arthus reaction. It is characterized by erythema, induration, oedema, haemorrhage and

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